

JUSTIN M. PARRA

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EDUCATION

Rutgers University – New Brunswick

Bachelor of Science in Electrical and Computer Engineering (ECE)

New Brunswick, NJ

Expected May 2027

Coursework: Principles of Electrical Engineering I & II, Microelectronic Circuits, Semiconductor Devices, Linear Systems and Signals, Control Systems Principles, Digital Logic Design, Computer Architecture, Embedded Systems & Microcontrollers.

TECHNICAL SKILLS

- **Languages:** C, C++ (C++11/17/20), Python, TypeScript, JavaScript, MATLAB, Verilog HDL, Assembly
- **Embedded & Microcontrollers:** ESP32-S3, STM32F405, STM32, ARM Cortex-M4, FreeRTOS, Bare-Metal C, HAL, Timer ISRs, PlatformIO
- **Hardware & PCB Design:** Altium Designer, KiCad 8.0, 4-Layer PCB Routing, Controlled Impedance, Coplanar Ground Shielding, Analog Signal Conditioning, Active Op-Amp Filters, RLC Transient Analysis, TVS Surge Protection, Decoupling
- **Protocols & Buses:** I2C (400 kHz), SPI, UART, RS-485, CAN Bus, PWM Motor Control, GPIO, ADC/DAC
- **Robotics & Control:** Cascade PID Control, Sensor Fusion (EMA, EKF Principles), Differential Drive Kinematics, Modified Flood Fill, A* Pathfinding, State Estimation
- **Tools & Platforms:** Linux (Arch, Ubuntu, WSL2), Git, GitHub, CMake, Make, GDB, LTspice, Multisim

ENGINEERING EXPERIENCE

Rutgers Micromouse

Lead Firmware & Hardware Engineer | Lead Controls Engineer | Co-President

New Brunswick, NJ

September 2024 – Present

- Engineered real-time embedded C/C++ firmware on ESP32-S3 and STM32 microcontrollers, utilizing FreeRTOS tasks to achieve deterministic 100 Hz sensor polling and sub-millisecond motor control response.
- Architected dual-loop cascade PID feedback controllers fusing 100 Hz IMU angular velocity (Bosch BNO055 9-DOF) with lateral distance error from ST VL53L1X Time-of-Flight sensors, eliminating motor drift and achieving sub-millimeter maze centering.
- Implemented memory-efficient Modified Flood Fill exploration and A* pathfinding algorithms in modern C++ with bit-manipulated state matrices, reducing autonomous traversal solve time by 20% under strict embedded SRAM limits.

LPT-KEYPAK

Engineering Intern

Tewksbury, NJ

June 2024 – August 2024

- Commissioned and validated automated high-speed packaging machinery for the Bill & Melinda Gates Foundation, boosting packaging line throughput by 30% during pilot production qualification runs.
- Diagnosed electromechanical and process failure points across automated manufacturing equipment using structured root-cause analysis (5-Whys, Pareto charts), reducing production defect rates by 15%.
- Interpreted 20+ industrial electrical schematics to troubleshoot PLCs, motor drives, and sensor relays; authored technical documentation and maintenance SOPs to standardize workflows.

VEX Robotics (Rutgers IEEE)

Lead Electrical Engineer

New Brunswick, NJ

September 2024 – Present

- Architected communication and distributed power delivery architectures for competition robotics, engineering RS-485 to UART to I2C transceiver networks between VEX Brain and microcontrollers.
- Designed schematics and selected components including differential transceivers, TVS transient surge diodes, decoupling filter networks, and bus termination resistors to protect sensitive logic from inductive spikes.
- Executed hardware-in-the-loop (HIL) stress testing and signal integrity validation under severe mechanical vibration using digital oscilloscopes to verify packet-loss rates <0.1% and guarantee deterministic bus arbitration.

TECHNICAL PROJECTS

Autonomous Micromouse Robot (Jerrieee) - 1st Place Regional Champion | C/C++, STM32F405RG, KiCad 8.0, 4-Layer ENIG PCB, DRV8847 2025

- Won 1st Place at Rowan Regional Competition; designed custom 4-layer ENIG PCB integrating 168 MHz ARM Cortex-M4 MCU, 5x optical ToF array over 400 kHz I2C, and dual DRV8847 motor drivers in an 84x76.5 mm chassis.
- Programmed real-time Flood-Fill maze navigation algorithm running in <850 μ s per cell transition with a 1 kHz cascaded PID control loop correcting gyro heading and velocity errors up to 4.2 m/s with zero integral windup.

Autonomous Vision Line-Following Robot | Embedded C++, ESP32-S3, OV2640 Camera, FreeRTOS, PlatformIO

2024

- Built an autonomous mobile robot using an ESP32-S3 and OV2640 camera module, developing an onboard vision-processing pipeline in C++ to detect track curvature in real time.
- Tuned closed-loop PID control algorithms for dual DC motors, maintaining trajectory tracking stability at speeds up to 1.2 m/s with zero track departures.